

Unbounded Orbit-Series and Reductivity Indices of Finite Quandles

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Abstract

For a finite quandle, let k , j , and i denote the least positive indices of local reductivity, trivialization of orbit series, and reductivity, respectively. We show that j cannot be bounded by a function of k , and i cannot be bounded by a function of j . For every $n \geq 3$, we construct a finite involutory quandle Q_n with $k(Q_n) = 2$ and exact value $j(Q_n) = \lfloor \log_3 n \rfloor + 1$. For every $m \geq 2$, we construct a finite quandle P_m with $(k(P_m), j(P_m), i(P_m)) = (2, 2, m)$. The first family uses finite label restrictions of the indexed Noce–Tracey–Traustason group, which is the $m = 2$ case of the broader family of Hadjievangelou–Traustason [3]; we derive the orbit-series filtration and exact cardinalities for these sections. The second family uses finite quotients of the reduced free group that preserve its exact nilpotency class; the same finite profile also follows from the known Cohen-group quotients [7]. The first family also yields an infinite involutory locally finite quandle in $\mathcal{LR}_2 \setminus \mathcal{OS}$, and we compute the exact sizes of the terms in its marked principal orbit series. All passages from groups to quandles, including the shift between the nilpotency class of the inner automorphism group and the reductivity index, are proved explicitly. This gives negative answers to both parts of Question 5.25 of Bonatto–Crans–Nasybullov–Whitney.

Keywords: finite quandle, local reductivity, orbit series, reduced free group, Engel element.

1 Introduction and main results

The classes $\mathcal{LR}_n$ and $t\mathcal{OS}_n$ were introduced in [1], where their relationship with the classes $\mathcal{R}_n$ was investigated. For finite quandles, these finite-depth conditions are equivalent at the level of the unions of the respective classes [1, Corollary 5.19]. Question 5.25 asks whether orbit-series depth can be bounded in terms of the local reductivity index, and whether the reductivity index can be bounded in terms of orbit-series depth. The numbering of the results in [1] used here follows arXiv version v3, dated August 18, 2020.

Recursive orbit decomposition was previously studied by Nelson–Wong [6, Section 5] under the name *subquandle depth*. Their procedure stops at connected subquandles, whereas j requires singleton terminal terms. For the nontrivial quandles Q_n below, all terminal connected subquandles are singletons, so these two depths agree.

For a quandle whose corresponding indices exist, set

$$\begin{aligned} k(Q) &= \min\{n \geq 1 : Q \in \mathcal{LR}_n\}, \\ j(Q) &= \min\{n \geq 1 : Q \in t\mathcal{OS}_n\}, \\ i(Q) &= \min\{n \geq 1 : Q \in \mathcal{R}_n\}. \end{aligned}$$

The classes are defined in Section 2.

Theorem 1.1. *For every integer $n \geq 3$, there is a finite involutory quandle Q_n such that*

$$k(Q_n) = 2, \quad j(Q_n) = \lfloor \log_3 n \rfloor + 1.$$

Theorem 1.2. *For every integer $m \geq 2$, there is a finite quandle P_m such that*

$$(k(P_m), j(P_m), i(P_m)) = (2, 2, m).$$

The quandle P_m can be chosen to have exactly m inner orbits, each of which is trivial as a quandle.

Corollary 1.3. *There is no function $f : \mathbb{N}_{>0} \rightarrow \mathbb{N}_{>0}$ such that $j(Q) \leq f(k(Q))$ for every finite quandle with defined indices. Nor is there a function g such that $i(Q) \leq g(j(Q))$ for all such quandles.*

Proof. The first family has $k = 2$ and unbounded j . The second has $j = 2$ and unbounded i . $\square$

For the first family, Proposition 3.2 computes the exact sizes of the terms in the marked principal orbit series. The infinite-product example in Section 3 is locally finite.

Theorems 1.1 and 1.2 are proved in Sections 3 and 4. The first family uses the indexed group of Noce–Tracey–Traustason [2]. Hadjievangeliou–Traustason [3] construct a broader family of locally finite 2-groups; their parameter $m = 2$ gives the same two-layer indexed group, and the groups G_n below are its finite restrictions to labels in $[n]$. The second family uses the reduced free group as treated by Darné [4]. The group-theoretic input results, with precise references, are collected in Subsections 3.1 and 4.1. The orbit-series filtration and exact cardinalities for the first finite sections, the preservation of class in a finite quotient, and the passages to quandles are proved in this paper.

2 Definitions and a reductivity criterion

We use the left convention for quandles [1]: a quandle is a nonempty set Q with an operation $\triangleright$ satisfying

$$a \triangleright a = a, \quad L_a : b \mapsto a \triangleright b \text{ is bijective}, \quad a \triangleright (b \triangleright c) = (a \triangleright b) \triangleright (a \triangleright c).$$

The group $\text{Inn}(Q) = \langle L_a : a \in Q \rangle$ is its inner automorphism group. We write $\text{Orb}(a, Q)$ for its orbit of a ; a quandle is connected if this group acts transitively. A quandle homomorphism preserves the operation, and an epimorphism here and below means a surjective homomorphism. Subquandles are also closed under inverse left translations. The trivial operation is given by $a \triangleright b = b$.

Write

$$T_n(a; c_1, \dots, c_n) = ((a \triangleright c_1) \triangleright \dots) \triangleright c_n$$

for the left-associated expression with n operations. The class $\mathcal{R}_n$ is defined by the identity

$$T_n(a; c_1, \dots, c_n) = T_n(b; c_1, \dots, c_n)$$

for all arguments; the class $\mathcal{LR}_n$ is defined by

$$T_n(a; \underbrace{b, \dots, b}_n) = b.$$

An orbit series is defined recursively by

$$Q_0 = Q, \quad Q_{r+1} = \text{Orb}(x_r, Q_r), \quad x_r \in Q_r.$$

At each step, the inner automorphism group of the current subquandle is used. The class $t\mathcal{OS}_n$ consists of quandles for which every such series reaches a singleton quandle no later than step n . The class $t\mathcal{OS}$ requires each series to reach a singleton quandle in finitely many steps, with the number allowed to depend on the series; set $t\mathcal{OS}_\omega = \bigcup_{n \geq 1} t\mathcal{OS}_n$. These definitions follow [1, Section 4.1]. Thus j measures the time to reach a singleton, rather than merely the time to stabilization. All three classes with index 1 coincide with the class of trivial quandles. Moreover, $Q \in t\mathcal{OS}_2$ if and only if each inner orbit of Q is trivial as a quandle; such an orbit may contain many elements.

For groups, we use the conventions

$$[a, b] = a^{-1}b^{-1}ab, \quad c^g = g^{-1}cg, \quad \gamma_1(H) = H, \quad \gamma_{r+1}(H) = [\gamma_r(H), H].$$

The trivial group has class zero. Double angle brackets denote normal closure; $\mathbb{F}_p$ denotes the field with p elements. Otherwise, exact class c means $\gamma_{c+1}(H) = 1$, $\gamma_c(H) \neq 1$.

Lemma 2.1. *Let S be a nonempty subset of a group, closed under conjugation by its own elements and their inverses. Then $a \triangleright b = a^{-1}ba$ makes S a quandle, and for $a \in S$*

$$\text{Orb}(a, S) = a^{\langle S \rangle}.$$

Proof. Idempotency and bijectivity are immediate, and

$$(a^{-1}ba)^{-1}(a^{-1}ca)(a^{-1}ba) = a^{-1}b^{-1}cba$$

proves self-distributivity. Left translations and their inverses are conjugations by elements of S and their inverses. They generate the action of $\langle S \rangle$, giving the orbit formula. $\square$

Proposition 2.2. *For every nonempty quandle Q and $n \geq 1$,*

$$Q \in \mathcal{R}_n \iff \gamma_n(\text{Inn}(Q)) = 1.$$

In particular, if $\text{Inn}(Q)$ has exact finite class c , then $i(Q) = c + 1$.

Proof. Set $\mathcal{L}(Q) = \{L_a : a \in Q\}$. Self-distributivity gives

$$L_{a \triangleright b} = L_a L_b L_a^{-1}.$$

Thus $a \mapsto L_a$ is an epimorphism of quandles when $\mathcal{L}(Q)$ is equipped with $s \triangleright t = sts^{-1}$. This direction of conjugation is opposite to the convention in Lemma 2.1.

For $n \geq 1$, the identity $\mathcal{R}_{n+1}$ for Q is equivalent to the identity $\mathcal{R}_n$ for $\mathcal{L}(Q)$: equality of the two expressions after the final operation for every final argument means equality of their left translations. Also, $Q \in \mathcal{R}_1$ if and only if $\mathcal{L}(Q)$ is a singleton. Indeed, if all L_a equal f , then idempotency gives $f(a) = a$ for every a . For the iterates $\mathcal{L}_0(Q) = Q$, $\mathcal{L}_{r+1}(Q) = \mathcal{L}(\mathcal{L}_r(Q))$, we obtain

$$Q \in \mathcal{R}_n \iff |\mathcal{L}_n(Q)| = 1.$$

Let $H = \text{Inn}(Q)$. The set $\mathcal{L}(Q)$ generates H and is preserved by conjugations in H , since $hL_a h^{-1} = L_{h(a)}$. The action of H on this set has image $\text{Inn}(\mathcal{L}(Q))$ and kernel $C_H(\mathcal{L}(Q)) = Z(H)$. Consequently,

$$\text{Inn}(\mathcal{L}(Q)) \cong H/Z(H).$$

For the upper central series $Z_0(H) = 1$, $Z_{r+1}(H)/Z_r(H) = Z(H/Z_r(H))$, induction gives

$$\text{Inn}(\mathcal{L}_r(Q)) \cong H/Z_r(H).$$

The quandle $\mathcal{L}_{n-1}(Q)$ is trivial if and only if $\mathcal{L}_n(Q)$ is a singleton; hence the latter condition is equivalent to $H = Z_{n-1}(H)$.

Finally, $H = Z_c(H)$ is equivalent to $\gamma_{c+1}(H) = 1$. This follows by induction on c , starting with $c = 0$: the condition $\gamma_{c+1}(H) = 1$ is equivalent to $\gamma_c(H) \leq Z(H)$, or $\gamma_c(H/Z(H)) = 1$. Applying induction to $H/Z(H)$ gives $H = Z_c(H)$. Taking $c = n-1$ proves the criterion, including $n = 1$. $\square$

Remark 2.3. The criterion corresponds to [1, Proposition 3.2], but is expressed by vanishing of a term of the series to distinguish a bound on the class from its exact value.

3 First family: unbounded depth

3.1 Group-theoretic input

We use the subgroup of the operator group over $\mathbb{F}_2$ from [2] generated by the generators with finite labels. Denote it by $\mathcal{G}$, and write its generators as

$$X = 1 + \text{ad}(x), \quad U_A = 1 + \text{ad}(u_A), \quad V_A = 1 + \text{ad}(v_A), \quad W_A = 1 + \text{ad}(w_A),$$

where A is a nonempty finite subset of $\mathbb{N}$. This same indexed group is the $m = 2$ member of the family of Hadjievangeliou–Traustason [3]. Their construction then has two indexed $v(i)_A$ -layers, and their $v(0)_A, v(1)_A, w_A$ correspond to our u_A, v_A, w_A , respectively. Their Lemma 3.2 gives the same commutator table, Proposition 3.3 gives a normal form, and Theorem 3.4 establishes the left 3-Engel property of X . Restricting labels to nonempty subsets of $[n]$ gives the finite subgroup G_n below, finite because the ambient group is locally finite. The space V^* and its product are constructed in [2, Section 2, before Lemma 2.2]; we use the right action $z \text{ad}(y) = z \cdot y$. All these generators are involutions. Each of the three families U, V, W generates an elementary abelian group. If $A \cap B = \emptyset$, put $A \sqcup B = A \cup B$; otherwise put $A \sqcup B = \emptyset$. By convention, $u_\emptyset = v_\emptyset = w_\emptyset = 0$, so $U_\emptyset = V_\emptyset = W_\emptyset = 1$. By [2, Lemma 2.3],

$$\begin{aligned} [U_A, V_B] &= U_{A \sqcup B}, & [V_A, W_B] &= W_{A \sqcup B}, \\ [W_A, U_B] &= V_{A \sqcup B}, & [W_A, X] &= U_A, \end{aligned} \tag{1}$$

and X commutes with all U_A, V_A . By [2, Proposition 2.4], every element has a unique normal form $X^\epsilon uvw$, where $\epsilon \in \{0, 1\}$ and u, v, w belong to the corresponding elementary abelian groups. Finally, the proof of [2, Theorem 2.5] establishes

$$[[X^g, X], X] = 1 \quad (g \in \mathcal{G}). \tag{2}$$

Every U_A is nontrivial. To see this in the original operator construction, choose a singleton B disjoint from A : the operator $\text{ad}(u_A)$ sends w_B to the nonzero $v_{A \cup B}$.

For $n \geq 1$, set

$$G_n = \langle X, U_A, V_A, W_A : \emptyset \neq A \subseteq [n] \rangle \leq \mathcal{G}, \quad [n] = \{1, \dots, n\}.$$

When a word is collected into normal form, new labels arise only through $\sqcup$, and therefore remain subsets of $[n]$. It follows that G_n is finite and

$$|G_n| \leq 2^{1+3(2^n-1)}. \tag{3}$$

Here the labels themselves are restricted, rather than only their sizes.

3.2 The filtration and normal closures

For an integer $s \geq 1$, define the subgroup

$$F_s = \langle X, U_A (|A| \geq s), V_A (|A| \geq 2s), W_A (|A| \geq 3s) \rangle \leq G_n,$$

using only labels in $[n]$. Its elements are precisely the normal forms with

$$u \in \langle U_A : |A| \geq s \rangle, \quad v \in \langle V_A : |A| \geq 2s \rangle, \quad w \in \langle W_A : |A| \geq 3s \rangle.$$

Indeed, the only correction terms in the collection process satisfy

$$\begin{aligned} [X, W_{\geq 3s}] &\subseteq U_{\geq 3s}, & [U_{\geq s}, V_{\geq 2s}] &\subseteq U_{\geq 3s}, \\ [W_{\geq 3s}, U_{\geq s}] &\subseteq V_{\geq 4s}, & [V_{\geq 2s}, W_{\geq 3s}] &\subseteq W_{\geq 5s}. \end{aligned}$$

The threshold does not decrease: disjoint labels give the sum of their sizes, whereas intersecting labels give the identity. Hereafter, $U_{\geq t}$ and the analogous symbols denote subgroups generated by labels of the corresponding size.

Lemma 3.1. *For every $s \geq 1$,*

$$\langle X^{F_s} \rangle = F_{3s}.$$

Moreover, $\langle X^{G_n} \rangle = F_1$.

Proof. First, we check that $F_{3s} \leq F_s$. In the table, rows correspond to generators of F_{3s} and columns to generators of F_s ; an entry indicates the layer containing a possible nontrivial commutator:

	X	$U_{\geq s}$	$V_{\geq 2s}$	$W_{\geq 3s}$
X	1	1	1	$U_{\geq 3s}$
$U_{\geq 3s}$	1	1	$U_{\geq 5s}$	$V_{\geq 6s}$
$V_{\geq 6s}$	1	$U_{\geq 7s}$	1	$W_{\geq 9s}$
$W_{\geq 9s}$	$U_{\geq 9s}$	$V_{\geq 10s}$	$W_{\geq 11s}$	1

All entries lie in F_{3s} . Since the generators are involutions, this check ensures invariance under conjugation by them in both directions. The subgroup contains X , so $\langle X^{F_s} \rangle \leq F_{3s}$.

For the reverse inclusion, set $K = \langle X^{F_s} \rangle$. It is normal in F_s and contains X . If $|A| \geq 3s$, then $W_A \in F_s$, so $[W_A, X] = U_A \in K$. Every label T of size at least $6s$ can be partitioned as $A \sqcup B$ with $|A|, |B| \geq 3s$; then $[W_A, U_B] = V_T \in K$. For $|T| \geq 9s$, take a partition with $|A| \geq 6s$, $|B| \geq 3s$. The previous step gives $V_A \in K$, and hence $[V_A, W_B] = W_T \in K$. These are all the generators of F_{3s} .

For the initial step, we check that $F_1 \leq G_n$. Possible nontrivial commutators between generators lie in the layers

$$\begin{aligned} [X, W_{\geq 1}] &\subseteq U_{\geq 1}, & [U_{\geq 1}, V_{\geq 1}] &\subseteq U_{\geq 2}, \\ [U_{\geq 1}, W_{\geq 1}] &\subseteq V_{\geq 2}, & [V_{\geq 2}, W_{\geq 1}] &\subseteq W_{\geq 3}, \\ [W_{\geq 3}, U_{\geq 1}] &\subseteq V_{\geq 4}, & [W_{\geq 3}, V_{\geq 1}] &\subseteq W_{\geq 4}, \\ [W_{\geq 3}, X] &\subseteq U_{\geq 3}. \end{aligned}$$

They all lie in F_1 ; the others are the identity or are obtained by inverting a commutator. Thus $\langle X^{G_n} \rangle \leq F_1$. Conversely, $[W_A, X] = U_A$ puts all U_A in the normal closure; partitions into two nonempty parts and $[W_A, U_B]$ give all V_T with $|T| \geq 2$; partitions with $|A| \geq 2$, $|B| \geq 1$ and $[V_A, W_B]$ give all W_T with $|T| \geq 3$. $\square$

3.3 The quandle and its depth

Proof of Theorem 1.1. Let $C_n = X^{G_n}$, and equip $Q_n = C_n$ with the operation $a \triangleright b = a^{-1}ba$. This is a finite quandle by Lemma 2.1. Since X is an involution and C_n consists of its conjugates, all elements of Q_n are involutions; hence Q_n is involutory. For $a = X^g$, $b = X^h$, conjugation by h^{-1} sends the pair to $X^{gh^{-1}}, X$. Thus (2) gives $[[a, b], b] = 1$ for all $a, b \in C_n$. Interchanging a, b gives $[[b, a], a] = 1$. Since $[b, a] = [a, b]^{-1}$, the commutator $[a, b]$ centralizes both generators. The subgroup $\langle a, b \rangle$ has class at most two. In particular, $a^{-1}ba$ commutes with b , so

$$(a \triangleright b) \triangleright b = b.$$

Hence $Q_n \in \mathcal{LR}_2$.

For $n \geq 3$, choose $|A| = 3$. Then $W_A \in F_1 = \langle C_n \rangle$ and $[W_A, X] = U_A \neq 1$. By Lemma 2.1, the orbit of X in Q_n is nontrivial, so Q_n is not trivial and $k(Q_n) = 2$. The index $j(Q_n)$ exists: by [1, Corollary 5.19], a finite quandle in $\mathcal{LR}_2$ belongs to $t\mathcal{OS}$, meaning that every orbit series reaches a singleton quandle. The length of such a series in a finite quandle is bounded: all inclusions before the singleton term are strict, since equality of consecutive terms would mean that the current term is connected and cannot decrease further. Therefore $Q_n \in \bigcup_{t \geq 1} t\mathcal{OS}_t$.

Choosing the anchor X at every step, set

$$S_0 = C_n, \quad N_r = \langle S_r \rangle, \quad S_{r+1} = X^{N_r}.$$

This is an orbit series by Lemma 2.1. Lemma 3.1 gives

$$N_r = F_{3^r} \quad (r \geq 0), \quad S_r = X^{F_{3^{r-1}}} \quad (r \geq 1).$$

If $r \geq 1$ and $3^r \leq n$, choose $|A| = 3^r$. Then $W_A \in F_{3^{r-1}}$, so $X, X^{W_A} \in S_r$; these elements are distinct because $[W_A, X] = U_A \neq 1$. Consequently, S_r is not a singleton. For $r = \lfloor \log_3 n \rfloor$, the series does not reach a singleton at any of the steps $0, \dots, r$, and $j(Q_n) \geq r + 1$.

For the matching upper bound, put $R = \lfloor \log_3 n \rfloor$. Then $3^R \leq n < 3^{R+1}$ and

$$S_{R+1} = X^{F_{3^R}}.$$

The group F_{3^R} has no generators W_A , since these would require $|A| \geq 3^{R+1} > n$. Its remaining generators X, U_A, V_A all commute with X , so $S_{R+1} = \{X\}$. We already know that S_R is non-singleton. Thus the principal series of X first reaches a singleton at step $R + 1$.

Every orbit series of Q_n is principal, by [1, Proposition 4.6 and Corollary 4.8]: finiteness implies stabilization, and the proposition identifies a stabilizing series with the principal series of any point in its intersection. Such a point y lies in $C_n = X^{G_n}$, so $y = X^g$ for some $g \in G_n$. Conjugation by g restricts to a quandle automorphism of Q_n and maps the entire principal series of X to that of y . Consequently all orbit series have the same first singleton step, and

$$\boxed{j(Q_n) = R + 1 = \lfloor \log_3 n \rfloor + 1.}$$

□

Proposition 3.2. *For the principal orbit series defined by $S_0 = C_n$ and $S_{r+1} = X^{\langle S_r \rangle}$, one has*

$$|S_r| = \sum_{t=3^r}^n \binom{n}{t} \quad (r \geq 0),$$

where an empty sum is zero. In particular, S_r is a singleton exactly when $3^r > n$.

Proof. For $r \geq 1$, the filtration lemma gives $S_r = X^{F_{3^{r-1}}}$. Put

$$\mathcal{W}_{s,n} = \{A \subseteq [n] : |A| \geq 3s\}.$$

By the unique normal form in Proposition 2.4 of [2], the W -component of an element of F_s is uniquely a product $w_T = \prod_{A \in T} W_A$ for some $T \subseteq \mathcal{W}_{s,n}$. The other components commute with X , so every element of X^{F_s} is X^{w_T} , and every such conjugate occurs.

The computation in the proof of Theorem 2.5 of [2] gives

$$X^{w_T} = 1 + \text{ad}(y_T), \quad y_T = x + \sum_{A \in T} u_A + \sum_{\{A, A'\} \subseteq T} v_{A \sqcup A'} + \sum_{\{A, A', A''\} \subseteq T} w_{A \sqcup A' \sqcup A''}.$$

Take $B = \{n + 1\}$ and apply $\text{ad}(y_T)$ to v_B in the ambient space V^* of [2]. Its U -component is

$$\sum_{A \in T} u_{A \cup B}.$$

The map $A \mapsto A \cup B$ is injective on subsets of $[n]$, and the corresponding basis vectors are linearly independent. Thus $X^{w_T} = X^{w_{T'}}$ implies $T = T'$. It follows that

$$|X^{F_s}| = 2^{|\mathcal{W}_{s,n}|} = 2^{\sum_{t=3s}^n \binom{n}{t}}.$$

Taking $s = 3^{r-1}$ proves the formula for $r \geq 1$. For $r = 0$, the same argument applied to $C_n = X^{G_n}$ uses all nonempty subsets of $[n]$, giving $|S_0| = 2^{2^n - 1}$, which is the stated formula. The singleton criterion follows because the exponent is zero exactly when $3^r > n$. $\square$

Corollary 3.3. *There is a locally finite infinite involutory quandle $Q \in \mathcal{LR}_2 \setminus \mathcal{OS}$.*

Proof. Let $Q = \prod_{n \geq 3} Q_n$ and $x = (X_n)_{n \geq 3}$, where X_n is the marked element in the n -th factor. The identity $(a \triangleright b) \triangleright b = b$ and involutivity hold coordinatewise, so $Q \in \mathcal{LR}_2$ and Q is involutory.

Consider the principal orbit series $P_0 = Q$, $P_{r+1} = \text{Orb}(x, P_r)$. For each projection $p_n : Q \rightarrow Q_n$, induction gives that $p_n(P_r)$ is the principal series term of Q_n anchored at X_n : a surjective quandle homomorphism maps each orbit onto the orbit of the image anchor. For any r , choose $n \geq \max(3, 3^r)$. The exact depth formula gives $p_n(P_r) \neq p_n(P_{r+1})$. Hence $P_r \neq P_{r+1}$ for every r , so this orbit series never stabilizes and $Q \notin \mathcal{OS}$.

The quandle Q is also locally finite. Let $q_1, \dots, q_d$ be any finite set of its elements. In each coordinate Q_n , these are d conjugates of X . By Proposition 2.9 and Corollary 2.7 of [2], the subgroup they generate in that coordinate is nilpotent of class at most $4d + 2$ and has exponent dividing 32. Consequently the subgroup generated by $q_1, \dots, q_d$ in $\prod_{n \geq 3} Q_n$ has the same class and exponent bounds. A finitely generated nilpotent group of finite exponent is finite: its lower central factors are finitely generated abelian groups of bounded exponent. The subquandle generated by $q_1, \dots, q_d$ lies in this finite subgroup, proving local finiteness. $\square$

Remark 3.4. Theorem 1.1 gives the exact orbit-series depth of this family. Minimal orders of quandles with prescribed indices are not determined here.

4 Second family: unbounded reductivity

4.1 The reduced free group and a finite quotient

Let $F(x_1, \dots, x_m)$ be the free group. Define Milnor's reduced free group [8] in the form of [4, Definition 1.2]:

$$RF_m = F(x_1, \dots, x_m) / \langle\langle [x_r, x_r^w] : 1 \leq r \leq m, w \in F(x_1, \dots, x_m) \rangle\rangle.$$

Each x_r commutes with all its conjugates. Darné identifies the free reduced quandle on m generators with the union of their conjugacy classes in RF_m [5, Proposition 7.10]. Together with the group facts below, this already gives the infinite analogue with indices $(k, j, i) = (2, 2, m)$: reducedness makes every orbit trivial, and for $m \geq 2$ the quandle is nontrivial, so $k = j = 2$. The action kernel on these classes is $Z(RF_m)$, so the inner group is $RF_m / Z(RF_m)$, of exact class $m - 1$. The construction below supplies finite quotients retaining the exact indices. The class bound goes back to [9, Lemma 1.3]; below we use its proof and the description of the center in [4]. By [4, Propositions 1.3 and 1.18],

$$\gamma_{m+1}(RF_m) = 1, \quad \gamma_m(RF_m) = Z(RF_m) \cong \mathbb{Z}^{(m-1)!}.$$

Here and below, the numbering in [4] follows arXiv version v2. Thus RF_m has exact class m . Choose $1 \neq z_m \in \gamma_m(RF_m)$.

We justify the existence of a finite quotient in which z_m survives. Consider the ring

$$\mathcal{A}_m = \mathbb{Z}\langle X_1, \dots, X_m \rangle / I,$$

where I is the two-sided ideal generated by words with at least one repeated letter. This is the reduced associative algebra of [4, Definition 1.4]; its additive basis consists of the empty word and all words without repetitions [4, Fact 1.5]. For the ideal $J = (X_1, \dots, X_m)$, we have $J^{m+1} = 0$; thus $1 + J$ is a group, with inverses

$$(1 + u)^{-1} = \sum_{t=0}^m (-u)^t.$$

By [4, Corollary 1.13], the Magnus map

$$\mu : RF_m \hookrightarrow 1 + J, \quad x_r \mapsto 1 + X_r$$

is injective. The expansion $\mu(z_m) - 1 = \sum_w c_w w$ has a nonzero integer coefficient c_w . Choose a prime p not dividing this coefficient and reduce the ring modulo p . Write $\bar{J}$ for the image of J in $\mathcal{A}_m/p\mathcal{A}_m$. The group $1 + \bar{J}$ is finite, since $\bar{J}$ has a finite basis over $\mathbb{F}_p$; its order is p^{d_m} , where

$$d_m = \sum_{t=1}^m \frac{m!}{(m-t)!}.$$

Let A_m be the image of RF_m in this group, and let $\phi_m : RF_m \twoheadrightarrow A_m$ be the epimorphism onto the image. Then $\phi_m(z_m) \neq 1$. Epimorphisms preserve the lower central series: $\phi_m(\gamma_t(RF_m)) = \gamma_t(A_m)$, as follows by induction from preservation of commutators. Hence A_m has exact class m .

To make the conjugacy classes of the chosen generators distinct, set

$$\alpha : RF_m \rightarrow (\mathbb{Z}/2\mathbb{Z})^m, \quad x_r \mapsto e_r, \quad G_m = \text{im}(\phi_m, \alpha) \leq A_m \times (\mathbb{Z}/2\mathbb{Z})^m.$$

Here e_r is the r -th standard basis vector. The map α is well-defined because all the relations lie in the commutator subgroup of the free group. Denote the epimorphism onto G_m by ψ_m , and the images of x_r by a_r . The group G_m is finite, is generated by $a_1, \dots, a_m$, and has exact class m : it is a quotient of RF_m , and its first coordinate preserves z_m . The abelian coordinates e_r are distinct, so a_r, a_s cannot be conjugate for $r \neq s$.

4.2 Orbits and the three exact indices

Proof of Theorem 1.2. Fix $m \geq 2$, and write $G = G_m$ for brevity. Set

$$C_r = a_r^G, \quad P_m = \bigcup_{r=1}^m C_r, \quad u \triangleright v = u^{-1}vu.$$

By Lemma 2.1, this is a finite quandle. The classes C_r are pairwise disjoint. The relations of the reduced free group give $[a_r, a_r^g] = 1$ for every $g \in G$. Any two elements a_r^g, a_r^h commute: conjugation by g^{-1} sends them to $a_r, a_r^{hg^{-1}}$. Thus every C_r is trivial as a quandle.

Consider the homomorphism

$$\rho : G \rightarrow \text{Sym}(P_m), \quad \rho(g)(v) = gvg^{-1}.$$

With the usual composition of permutations, this is a homomorphism; the maps $v \mapsto g^{-1}vg$ form an antihomomorphism. We have $L_u = \rho(u^{-1})$. Since P_m contains all the a_r , it generates G ; hence $\text{Inn}(P_m) = \rho(G)$. Its orbits are exactly $C_1, \dots, C_m$. Every orbit series reaches one of the C_r at its first step and a singleton quandle at its second. Thus $P_m \in t\mathcal{OS}_2$. The quandle is not trivial: otherwise all the a_r would commute pairwise, and the group G that they generate would be abelian, contrary to its class $m \geq 2$. Therefore $j(P_m) = 2$.

The element $u \triangleright v$ lies in the same C_r as v , and therefore commutes with v . It follows that $(u \triangleright v) \triangleright v = v$, so $P_m \in \mathcal{LR}_2$. Nontriviality gives $k(P_m) = 2$.

The kernel of ρ is $C_G(P_m) = Z(G)$, since P_m generates G . Consequently,

$$\text{Inn}(P_m) \cong G/Z(G).$$

The quotient $G/Z(G)$ has exact class $m - 1$. Indeed, $\gamma_m(G) \leq Z(G)$, so $\gamma_m(G/Z(G)) = 1$. If $\gamma_{m-1}(G/Z(G)) = 1$, then $\gamma_{m-1}(G) \leq Z(G)$, forcing $\gamma_m(G) = 1$, a contradiction. Proposition 2.2 now gives $i(P_m) = m$. All the required equalities follow. $\square$

Remark 4.1. The existence assertion of Theorem 1.2 also follows from known finite Cohen groups. For an odd prime p , let

$$K_m^{\mathbb{Z}/p} = RF_m / \langle\langle x_1^p, \dots, x_m^p \rangle\rangle.$$

Cohen–Mikhailov–Wu identify K_m with RF_m and prove that $K_m^{\mathbb{Z}/p}$ has exact nilpotency class m [7, Sections 1–2]. This group is finite: it is finitely generated and nilpotent, with finite abelianization $(\mathbb{Z}/p\mathbb{Z})^m$. Indeed, its lower central factors are generated by commutators in the finite abelianization and hence are finite. The generator images are nonconjugate because their abelianization classes are distinct. Each generator commutes with all its conjugates by the reduced relations. Writing $\bar{x}_r$ for the generator images, set

$$P = \bigcup_{r=1}^m \bar{x}_r^{K_m^{\mathbb{Z}/p}}.$$

These m conjugacy classes form exactly m trivial inner orbits of P . The argument above gives $\text{Inn}(P) \cong K_m^{\mathbb{Z}/p}/Z(K_m^{\mathbb{Z}/p})$ of class $m - 1$, and hence $(k(P), j(P), i(P)) = (2, 2, m)$. Thus this finite-profile existence statement is already a consequence of the known group construction; the preceding Magnus-quotient proof gives another explicit realization.

Remark 4.2. The abelian factor $(\mathbb{Z}/2\mathbb{Z})^m$ is needed to ensure exactly m distinct orbits. For unboundedness of i with $j = 2$, A_m itself is sufficient. The prime p is chosen after choosing z_m ; we do not assert that an arbitrary z_m survives for any prime p fixed in advance.

4.3 An example for $m = 2$

To illustrate the index shift, take the group of upper unitriangular matrices $G = UT_3(\mathbb{F}_2)$. Writing E_{rs} for a matrix unit, choose generators

$$a = 1 + E_{12}, \quad b = 1 + E_{23}, \quad z = 1 + E_{13}.$$

This group has order 8, class 2, and center $\{1, z\}$. The generator classes are $\{a, az\}$ and $\{b, bz\}$. On $P = \{a, az, b, bz\}$, the operation has table

$\triangleright$	a	az	b	bz
a	a	az	bz	b
az	a	az	bz	b
b	az	a	b	bz
bz	az	a	b	bz

There are two trivial orbits of size 2, and $\text{Inn}(P) \cong G/Z(G)$ is nontrivial and abelian. Thus $(k(P), j(P), i(P)) = (2, 2, 2)$.

5 Scope of the results

Both obstructions occur already at the least nontrivial value of the controlling index, namely 2. The first family consists of a single group conjugacy class; the second consists of a union of the classes of chosen generators. It is not the full conjugation quandle on the entire group.

Consequently, [1, Corollary 5.24], which concerns the full quandle $\text{Conj}(G)$, does not bound the class of the group in the construction of P_m .

The finite quotient in the second family is chosen to preserve a nonzero commutator of weight m . An arbitrary finite quotient may decrease the class and does not ensure the required index. The proofs establish the existence of these families and the exact orbit-series depth and term cardinalities of the first family. Optimal orders are not determined here.

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